

Committee Management Web Application

Updated Project Documentation — Version 2

Flask + SQLAlchemy + WTForms + Jinja + Turso + Render

1. What Changed Since the Last Version

This version replaces the earlier draft with two sets of updates:

- **Corrected payment rule: the contribution amount is fixed per committee. Every member pays the same amount — it is not set individually per member.**
- **Expanded scope: the app now supports multiple committees running at the same time, each either weekly or monthly, with a fuller page structure (public pages, an admin dashboard with cards, and an “Explore” management screen).**

2. Overview

The application manages one or more rotating committees (ROSCA / kameti). In each committee, every member pays the same fixed amount every period (a week or a month, depending on the committee's type). At the end of each period, one member — chosen outside the app, by the owner or by vote — receives the full pool collected that period. A member can only receive the payout once per committee. A committee's number of periods equals its number of members, since it ends once everyone has received the payout exactly once.

3. Key Business Rules

- Contribution amount is fixed at the committee level — every member in that committee owes the same amount each period.
- A single admin (or small admin team) can run several committees at once, each independent of the others.
- Each committee is either Weekly or Monthly — set once, when the committee is created.
- Total periods in a committee = total number of members in that committee (one payout per period, no repeats).
- Members can only be added or removed before the committee's start date. After that date, the member list is locked — only details like a member's name can still be edited.
- A committee is automatically marked “completed” once every member has received the payout once.

4. Who Uses the App

There are two audiences:

- General visitors / committee members — no login required. They can view every committee, every member, and the full payment/payout history. This is read-only.
- The Admin — logs in, and is the only one who can add, edit, or remove members, record payments, record payouts, and create new committees.

5. Page-by-Page Walkthrough

Page / Route	Who Can See It	What It Shows
Home (/)	Everyone — no login	Every committee running right now, each with its current period (this month or this week, depending on the committee's type) and a list of members who have already received the payout in that committee.
Committee Page (/committee/<id>)	Everyone — no login	One committee's status: total periods (equal to the total number of members), which period it is now, and the full member list with each member's current-period status (paid / not paid / already received). Clicking a member opens their page.
Member Page (/member/<id>)	Everyone — no login	One member's complete record: status for every past period, total amount contributed so far, whether and when they received the payout, and a pie chart showing their contribution as a share of the committee total.
Admin Login (/auth/login)	Admin only	Username and password entry point.
Browser Verification (/auth/verify)	Admin only, first time on a new browser	A 6-digit code is emailed to the admin's registered email. Entering it correctly marks this browser as trusted for future logins.
Admin Dashboard (/admin)	Admin only	Every committee shown as a card (active and completed). Each card has an “Explore” button, and there is a “Create New Committee” button at the top.
Explore (/admin/committee/<id>/explore)	Admin only	Full member list for one committee with editable status per member (mark received / pending for the current period). Controls at the top: Add Member, Edit Member, Remove Member.
Create Committee (/admin/committee/new)	Admin only	A form to start a new committee: name, start date, frequency (weekly or monthly), contribution amount, and member count.

5.1 Home Page

The landing page for every visitor. It lists every committee currently running, each shown with its current period label (“This Month” or “This Week”, matching the committee's frequency) and the members who have already received the payout so far in that committee. Clicking a committee opens its dedicated page.

5.2 Committee Page

Reached by clicking a committee from the Home page (URL pattern /committee/<id>). Shows that committee's overall status, its total number of periods (equal to its member count), and the full member list with each member's status for the current period. Clicking any member opens their individual page.

5.3 Member Page

Reached by clicking a member (URL pattern /member/<id>). Shows that member's complete history — their status in every period so far, their total amount contributed to date, whether they've received the committee payout (and when, if so), and a simple pie chart showing their contribution as a percentage of the committee's total collected so far.

5.4 Admin Login & Browser Verification

The admin logs in with a username and password. If logging in from a browser that has never been verified before, a 6-digit code is emailed to the admin's registered email address; entering it correctly marks that browser as trusted so future logins from it skip this step. This protects the admin panel even if the password alone were guessed.

5.5 Admin Dashboard

Visible only after login. Every committee (active and completed) is shown as a card. A “Create New Committee” button starts the form described in section 5.7. Each card has an “Explore” button leading to that committee's management screen.

5.6 Explore (Committee Management Screen)

The admin's working screen for one committee. It lists every member with their current-period status, which the admin can change directly (e.g. mark as “received” or “pending”). At the top are Add Member, Edit Member, and Remove Member controls.

Important: if the admin tries to add or remove a member after the committee's start date, the app shows a warning — “Cannot add or remove a member until the current cycle is completed” — and blocks the action. Editing a member's existing details (such as correcting their name) remains allowed at any time.

5.7 Create New Committee

A form the admin fills out to start a new committee. Once submitted, the committee is created and the admin can begin adding members. Adding and removing members is only possible up until the start date entered here.

Field	Notes
Committee Name	Free text, e.g. “Office Committee — 2026”.
Start Date	The date the committee begins. Once this date is reached, members can no longer be added or removed — only edited (e.g. correcting a name).
Frequency	Dropdown: Weekly or Monthly. This decides whether each period on the committee's page is labeled by week or by month.
Contribution Amount	One fixed amount that every member pays each period. This is set once at the committee level, not per member.
Member Count	Either set a target number of members now, or leave it open and let it lock in automatically to however many members have been added by the start date. (Flagged below as a decision to confirm.)

6. Data Model Summary

The database only stores plain values (text, numbers, dates, true/false flags) — no uploaded files or images.

Record Type	What It Stores
Committee	Name, frequency (weekly or monthly), fixed contribution amount, start date, status (active/completed), total number of periods (equal to member count).
Member	Name, gender, which committee they belong to, whether they've received the payout yet, which period they received it in, date joined.
Payment	One record per member per period: expected amount (the committee's fixed amount), whether it was actually paid, and the date paid. Kept as its own record even though the amount is fixed, so every individual transaction is auditable.
Payout	One record per period per committee: which member received the pool, the total amount, and the date.
Admin	The administrator's login username, email, and password (stored securely, never in plain text).
Trusted Browser	A record created after a browser passes email verification once, so the admin isn't asked to re-verify every login from that same browser.
Verification Code	A short-lived, one-time 6-digit code used only during new-browser login.

7. Application Flow, Start to Finish

- Admin logs in (with browser verification the first time on a given device).
- Admin creates a new committee: name, start date, frequency, fixed contribution amount, and member count.
- Before the start date, admin adds all members to that committee via Explore.
- Once the start date passes, the member list locks — no more adding or removing.
- Each period (week or month), members pay the fixed amount; admin marks each member's payment status in Explore.
- At period end, one member is chosen (outside the app) to receive the pool; admin records this as the payout for that period.
- The chosen member's status updates to “received” and they are excluded from future payout selection in that committee.
- This repeats until every member has received the payout once, at which point the committee is automatically marked completed.
- At any time, the public Home, Committee, and Member pages reflect the latest status for anyone to view.

8. Technology Stack

- Backend: Python + Flask
- Database & ORM: SQLAlchemy connected to Turso (a hosted, persistent SQLite-compatible database)
- Forms & security: Flask-WTF (WTForms + CSRF protection)
- Frontend: Jinja2 templates with HTML/CSS

- Authentication: Flask session login + email-based new-browser verification
- Hosting: Render (free web service) for the app, Turso (free tier) for the database

9. Open Items to Confirm With the Client

- Member Count on committee creation: should the admin fix the number upfront, or should it be determined automatically by however many members get added before the start date?
- Timezone for period labels and history timestamps (local time vs. UTC).
- Gender field: free text, or a fixed list of options?
- When a Weekly committee's "current period" changes — does the week start on a fixed day (e.g. every Monday), or exactly 7 days after the committee's start date?